

Math317

Numerical Methods

Course Overview



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What Are Numerical Methods

- Techniques by which mathematical problems are formulated so that they can be solved with arithmetic operations $\{+,-,*,./\}$ that can then be performed by a computer.

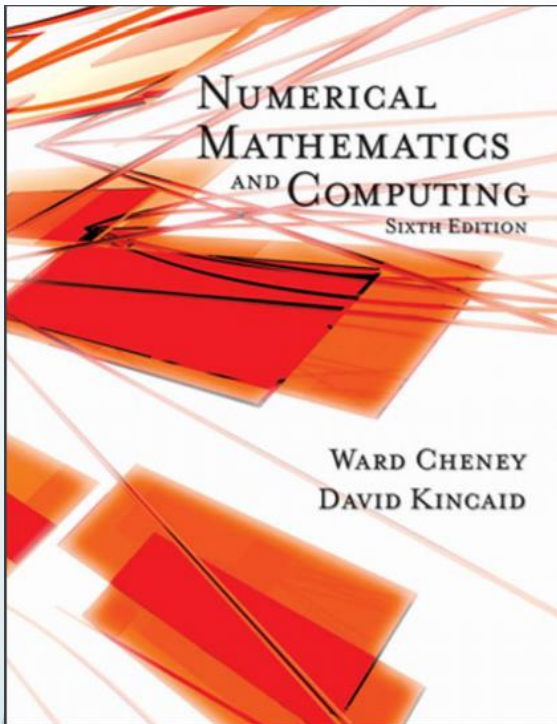
About this Course

- This course is intended for teaching numerical methods for computer science students at the senior level.
- Topics:
 - Errors in Computation.
 - Taylor Series representation of a function.
 - Representation of Numbers and Floating point representations.
 - Finding Roots of Equations.
 - Numerical Differentiation.
 - Numerical Integration.
 - Interpolation and Splines for curve fitting.
 - Regression.
 - Numerical methods for Systems of equations.
 - Numerical method for computing of Eigen-values and Eigen-vectors.

Goals of a Course?

- The Main goal of this course is to help the student in understanding some of the many methods for solving typical problems that arise in science, engineering, and technology on a modern computer.

Textbook



- Textbook (optional): Numerical Mathematics and Computing by Ward Cheney and David Kincaid.
- Textbook not a substitute for lectures:
 - Class presentation may not follow book.
 - Skip many chapters and cover extra material.

- *Additional sources*

- [Numerical Methods for Engineers](#), 6/e, by Steven C. Chapra and Raymond P. Canale. -- McGraw-Hill Higher Education 2010. ,
- <http://numericalmethods.eng.usf.edu>

Requirements?

- Weekly written homework assignments.
- Midterm exam: in-class, closed-book.
- Scheduled for week 8.
- Final exam on week 16.
- No make-up exams given unless you have serious, documented medical emergency.

Grading?

- Final exam: 50% of final grade.
- Midterm: 30% of final grade
- Homework + QUIZES + Class Participations + Projects : 20% of final grade

Why we need numerical Methods?

- Numerical methods are extremely powerful problem-solving tools.
- During your career, you may often need to use commercial computer programs (canned programs) that involve numerical methods. You need to know the basic theory of numerical methods in order to be a better user.
- You will often encounter problems that can't be solved by existing canned programs; you must write your own program of numerical methods.
- Numerical methods are an efficient vehicle for learning to use computers.
- Numerical methods provide a good opportunity for you to reinforce your understanding of mathematics.

Course outcomes?

Upon completion of this course, the student should be able to :

- Explain the role and the limitations of the computer in solving mathematical and engineering problems.
- Analyze a computation for error and discuss the types and sources of errors involved.
- Explain how error accumulates and discuss the errors inherent in using standard floating point numbers.

Course outcomes?

- Implement mathematical algorithms to:
 - Evaluate functions,
 - Find approximate roots of equations,
 - Solve systems of linear equations ,
 - Perform numerical differentiation and integration of single and Multiple integration: double and triple integrals applications.
 - Vector fields; gradient, divergence, curl, and the del operator, Fit a curve to a set of data, Least square methods.

